

# Anchored Conditionally-Gated Control for Wheeled Bipedal Balance and Disturbance Recovery

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**Abstract**—Wheeled bipedal robots face a fundamental trade-off: stiffness for sub-millimeter precision shrinks push-recovery capture, while integral action that cancels equilibrium bias winds up during disturbances. We present the Anchored Conditionally-Gated Controller (ACC), a two-channel torque assembly whose core mechanism is a proximity-gated anchor: a position integral confined to a 15 cm neighborhood, enabled by an asymmetric envelope follower ( $\tau_{\text{attack}} \approx 23$  ms,  $\tau_{\text{release}} \approx 1.5$  s) that distinguishes quiet stance from post-push ringdown. ACC delivers omnidirectional push recovery ( $F_{\text{min}}=83$  N,  $F_{\text{med}}=117$  N, 8-direction sweep,  $N=10$ ) and sub-millimeter idle precision—under idealized conditions (clean sensors, 0 ms delay, stiff contact, MuJoCo),  $0.73 \pm 0.07$  mm CoM RMS ( $N=10$ , 95% CI [0.68, 0.78] mm); degrades to  $\sim 1.2$  mm (tire compliance), 2.30 mm (10 ms delay), and hardware-projected 1–3 mm (encoders, backlash, stiction, IMU noise).  $F_{\text{max}}$  bifurcates from 96 N to 47 N at any non-zero delay. Six classical baselines—four 4-state TWIP variants (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo and direct-torque)—all fail (100 % fall rate,  $N=20$ –60); a preliminary PPO baseline (5.4M steps, single seed) suffers an 85 % fall rate over multi-height transitions; removing the PID servo path (4-state Direct Torque LQR) improves survival  $\sim 38$  % but does not prevent the fall. Flight recovery and per-leg terrain adaptation are demonstrated ( $N=10$  per condition); core claims rest on  $N \geq 5$ .

**Index Terms**—Wheeled biped robot, conditionally-gated control, anchored standing, proximity-gated integral, push recovery, flight recovery, terrain adaptation.

## I. INTRODUCTION

Wheeled biped robots need millimeter-level standing precision, omnidirectional push recovery, and terrain adaptation—but no single classical controller integrates all three: LQR handles balance [1], WBC posture [2], MPC terrain [5], and RL locomotion [25].

The difficulty is fundamental: lowering  $k_p$  widens the capture basin at the cost of balance stiffness; omitting integral action leaves a 1.3 Nm equilibrium bias sustaining a persistent 55 mm p-p limit cycle (RMS  $\approx 39$  mm) regardless of  $k_p$ —yet a global integral winds up during pushes.

Preliminary PPO training (5.4M steps, single seed, in our BalanceEnv) showed limited height generalization: the policy stood at a single height (Surv  $\approx 3.20$  s, vs. 0.52 s for LQR) but failed to generalize across 0.40 m to 0.70 m—height RMSE  $> 50$  mm, 85 % falls on squat transitions—consistent with known difficulty of model-free RL on coupled pitch–height dynamics [33].

We present the Anchored Conditionally-Gated Controller (ACC), addressing both dilemmas through *conditional activation*. The core mechanism is a **proximity-gated anchor**:

a position integral confined to a 15 cm neighborhood, enabled by an **asymmetric envelope follower** ( $\tau_{\text{attack}} \approx 23$  ms,  $\tau_{\text{release}} \approx 1.5$  s) that distinguishes quiet stance from post-push ringdown. Coupled with a **gated damping boost**, ACC achieves  $0.73 \pm 0.07$  mm CoM RMS ( $N=10$ ) under idealized conditions (clean sensors, 0 ms delay) and omnidirectional push survival ( $F_{\text{min}}=83$  N,  $F_{\text{med}}=117$  N)—sub-millimeter precision at equivalent push performance to the P-only baseline (no integral action), a  $\sim 53\times$  RMS improvement. Idle precision requires both the spatial proximity gate and velocity-envelope gating: the gate alone (L2.5, 49.8 mm RMS) sustains a limit cycle indistinguishable from the global integral (L2), while the envelope follower provides the temporal gating that breaks the cycle; S2 (no boost) and S3 (symmetric EMA) both retain sub-millimeter idle precision (0.73 mm), whereas S4 (no proximity gate) windups—confirming that idle precision requires the proximity gate plus any velocity-envelope gating, but neither the asymmetric envelope specifically nor the damping boost; the asymmetric envelope and boost are essential only for post-push ringdown (S2, S3 ringdown = never; Table IV). Six classical baselines—four 4-state TWIP variants (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo and direct-torque)—all fail (100 % fall rate), confirming that platform complexity, not servo lag alone, dominates the failure. Preliminary model-free PPO (5.4M steps, single seed) suffers an 85 % fall rate; removing the PID servo path (Direct Torque LQR) yields only a  $\sim 38$  % improvement with persistent 100% falls.

The main contributions are:

- 1) A **proximity-gated anchor** with asymmetric envelope follower achieving sub-millimeter idle precision without sacrificing omnidirectional push survival.
- 2) A **two-channel conditionally-gated torque assembly** with flight recovery and per-leg terrain adaptation as independent extensions ( $N=10$ ).
- 3) A **factorial ablation** (additive L0  $\rightarrow$  L3 + subtractive S0  $\rightarrow$  S5) with six classical baselines (four 4-state-derived and two 6-state coupled) and a preliminary PPO baseline (5.4M steps, single seed).

## II. RELATED WORK

### A. Wheeled Bipedal Platforms and Classical Control

Ascento [1] demonstrated two-wheeled jumping with LQR balancing, later extended to WBC with kinematic loops [2]. Handle [19] pioneered wheeled-leg coordination;

Swiss-Mile [20] combined driving with quadrupedal walking. BITeno [21], DIABLO [3], Whleaper [4] (closest to our morphology, uses LQR+PPO residual architecture), and SKATER [5] applied LQR, PPO, and MPC for balancing and locomotion. Whleaper’s LQR+PPO hybrid is the closest prior structured-controller-plus-learned-residual architecture; ACC targets the same paradigm with a stronger classical prior. Zamani et al. [31] demonstrated coupled pitch-roll LQR ( $\theta_{\text{RMS}} \approx 1.5^\circ$ ,  $\phi_{\text{RMS}} \approx 2.0^\circ$ ) on a rigid two-wheeled platform without articulated legs, where torque commands act directly on wheels—a structurally simpler control problem with fewer unstable modes and no leg-height dynamics. Our 6-state LQR replicates their coupled structure on a legged morphology and fails; the 4-state Direct Torque LQR—removing the servo path and re-optimizing roll gain ( $k_{p,\text{roll}} = 55.0$ )—also fails (100% fall rate, 0.72 s). Both LQR variants use gains derived from a simplified 4-state TWIP model, not a full 10-DOF linearization of the articulated-leg plant (see Limitation 7); these results are therefore empirical—demonstrating that the specific LQR implementations tested fail—not a proof that LQR is structurally insufficient. Two facts—Zamani succeeds on a rigid two-wheeler, both our LQR variants fail on articulated legs—indicate that platform complexity (articulated legs, height dynamics), not servo lag alone, is the dominant factor differentiating Zamani’s result. Cascade control [15], [23] and WBC [14] use fixed-gain nested loops without conditional gates; gain scheduling [18] varies parameters continuously with one variable rather than responding to qualitatively distinct physical regimes. We additionally benchmarked a task-priority QP whole-body controller both standalone and as an assist layer on ACC (§V-E); it fails to resolve the precision-recovery trade-off because its fixed task-priority formulation lacks conditional regime-gating. Split-range, selector, and override architectures [16], [17] already implement continuous blending in process control. ACC adopts this principle; its contribution is the synthesis of physically-motivated gating surfaces (spatial proximity, temporal velocity envelope, pitch safety-interlock) derived from wheeled-biped failure modes rather than generic process-variable limits. ACC’s conditional gates connect to hybrid/switched systems [35], where continuous dynamics are partitioned by discrete switching surfaces; smoothstep softens these to  $C^1$  transitions. A recent review [29] surveys wheel-legged biped platforms and control approaches, noting the absence of a unified classical solution.

#### B. Disturbance Rejection, Anti-Windup, and ACC’s Distinction

DOB [10], [11], [32] and ADRC [9], [12] provide temporal disturbance estimation but cannot distinguish a push from post-push ringdown—both occupy the same frequency band. Saturation-triggered conditional-integration AW triggers on actuator saturation; ACC’s anchor uses **spatial, physics-conditioned gating**: the proximity gate confines integration to a 15 cm neighborhood, and the asymmetric envelope distinguishes quiet stance from ringdown via velocity envelope—replacing the estimation-speed-vs-noise trade-off with a time-

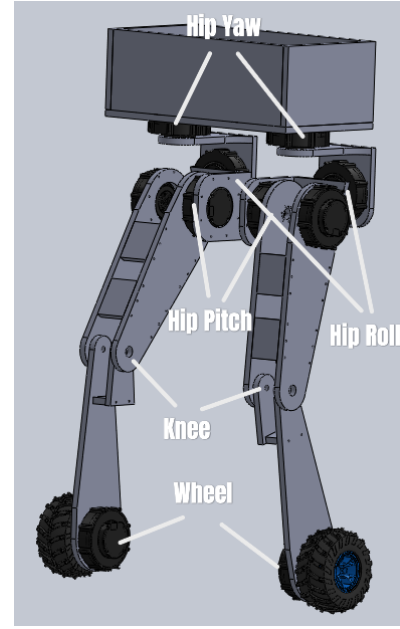


Fig. 1. Robot joint layout with world coordinate frame (X: forward, Y: left, Z: up, right-handed). The CoM is located at the torso geometric center. Each leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), wheel (WH).

domain attack-vs-release trade-off. Push recovery via capture-point [8], [26] and flight-phase control [7] is complementary. Blind stair climbing via RL was shown on Ascento [6]. Terrain adaptation uses exteroceptive [27] or proprioceptive [13], [28] sensing; ACC extends proprioceptive contact-point estimation to wheeled bipeds. Gated control for humanoid balancing was explored by Hyon et al. [34]; ACC extends conditional gating to wheeled bipeds with continuous smoothstep transitions.

### III. ROBOT PLATFORM

#### A. Robot Morphology and Simulation

The robot (Fig. 1) is a ten-DOF wheeled biped with five actuated joints per leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), and drive wheel (WH). Mass: 8.1 kg; thigh: 0.26 m; shin: 0.28 m; wheel radius: 0.06 m; hip width: 0.23 m. Torque limits: 30 Nm (HR, HY, WH), 150 Nm (HP, KN). MuJoCo [30] simulation at 500 Hz ( $\Delta t_{\text{sim}} = 0.002$  s). ACC controller: 100 Hz ( $\Delta t_{\text{ctrl}} = 0.01$  s); PPO and LQR baselines: 50 Hz (10 physics substeps). All experiments use 400 Nm/s torque rate limit, raw torque commands direct to actuators. MuJoCo integrator: implicitfast (4 iterations, 8 linesearch iterations); cone: pyramidal (elliptic for wheels,  $\mu_s = 1.2$ ,  $\mu_k = 0.8$  body); armature:  $0.008 \text{ kg}\cdot\text{m}^2$  (wheels),  $0.02 \text{ kg}\cdot\text{m}^2$  (leg joints); contact stiffness/damping:  $\text{solref} = \{0.002, 1\}$  (body),  $\{0.005, 0.7\}$  (wheels). The stiff body contact ( $\text{solref} = \{0.002, 1\}$ ) idealizes a rigid ground. Softening  $\text{solref}[0]$  to 0.02 (tire-on-asphalt) increases idle RMS  $\sim 60\%$  to  $\sim 1.2$  mm, so the 0.73 mm claim is conditional on stiff ground contact; tire compliance adds 0.1 mm to 0.5 mm on real surfaces (see §VI).

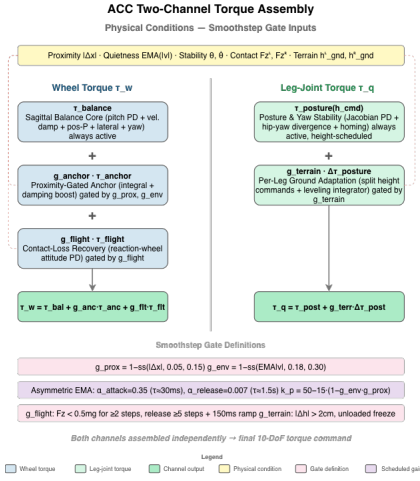


Fig. 2. ACC two-channel torque assembly. Physical conditions gate each component via continuous smoothstep functions (dashed).

#### IV. ACC FORMULATION

##### A. Two-Channel Torque Assembly

ACC assembles a PD balance core, proximity-gated anchor integral, and posture regulation into two torque channels—wheel torque ( $\tau_w \in \mathbb{R}^2$ ) and leg-joint torque ( $\tau_q \in \mathbb{R}^8$ )—with flight and terrain as independent extensions (Fig. 2):

$$\begin{aligned}\tau_w &= \tau_{\text{balance}} + g_{\text{anchor}} \tau_{\text{anchor}} + g_{\text{flight}} \tau_{\text{flight}}, \\ \tau_q &= \tau_{\text{posture}}(h^{\text{cmd}}) + g_{\text{terrain}} \Delta \tau_{\text{posture}}(h^{\text{cmd},L}, h^{\text{cmd},R}).\end{aligned}\quad (1)$$

Each gate  $g_*$  except the flight gate is a smoothstep function<sup>1</sup> of physical conditions (proximity, quietness, contact, terrain disparity)—not a discrete switch. The flight gate  $g_{\text{flight}}$  uses discrete load-based hysteresis with asymmetric engage/release timers (2-step engage, 5-step release) to prevent re-engaging balance torque during landing bounce. Wheel torques control sagittal balance and yaw; leg-joint torques control posture and per-leg height adaptation.

“Conditionally-gated” denotes smoothstep activation by physical conditions ( $g_{\text{prox}}$ ,  $g_{\text{env}}$ ,  $g_{\theta}$ ,  $g_{\text{terrain}}$ ). Continuous blending is standard in split-range control [16], [17]; ACC’s distinction is the specific gating surfaces—spatial proximity, temporal velocity envelope, pitch safety-interlock—chosen for wheeled-biped failure modes, not generic process variables. Distinct from cascade control [15]. Unlike industrial override (selector among parallel controllers by process-variable limits), ACC gates components within one assembly by physical regime—spatial (proximity), temporal (envelope), safety-interlock (pitch)—not control-error thresholds.

##### B. Sagittal Balance Core ( $\tau_{\text{balance}}$ )

The sagittal balance component is always active and acts entirely in the wheel-torque channel:

$$\tau_{\text{balance}} = \begin{bmatrix} \tau_{\text{com}} + \tau_{\text{diff}} \\ \tau_{\text{com}} - \tau_{\text{diff}} \end{bmatrix}, \quad \begin{aligned} \tau_{\text{com}} &= \tau_{\text{pitch}} + \tau_{\text{damping}} + \tau_{\text{pos}}, \\ \tau_{\text{diff}} &= \tau_{\text{lat}} + \tau_{\text{yaw}}. \end{aligned}\quad (2)$$

<sup>1</sup>smoothstep( $x, a, b$ ) =  $3t^2 - 2t^3$  with  $t = \text{clip}((x - a)/(b - a), 0, 1)$ . Smoothstep itself is  $C^1$ ; composed with the  $C^0$  asymmetric EMA (Eq. 12), composite  $g_{\text{env}}$  is  $C^0$ .

where  $\tau_{\text{com}}$  and  $\tau_{\text{diff}}$  are common-mode and differential components, respectively. The individual terms are:

$$\tau_{\text{pitch}} = k_p \cdot \theta + k_d \cdot \dot{\theta}, \quad (3)$$

$$\tau_{\text{damping}} = -k_{\text{vel}} \cdot v_{\text{sag}}, \quad (4)$$

$$\tau_{\text{pos}} = -k_{\text{pos}} \cdot \text{clip}(\Delta x, \pm 0.10 \text{ m}), \quad (5)$$

$$\tau_{\text{lat}} = k_{p,\text{roll}} \cdot \phi + k_{d,\text{roll}} \cdot \dot{\phi}, \quad (6)$$

$$\tau_{\text{yaw}} = k_{p,\text{yaw}} \cdot e_{\psi} + k_{d,\text{yaw}} \cdot \omega_z, \quad (7)$$

where  $\theta$  is the torso pitch angle,  $\dot{\theta}$  is the pitch rate,  $v_{\text{sag}}$  is the sagittal body velocity,  $\Delta x$  is the sagittal position error from the latched home position,  $\phi$  is the roll angle, and  $e_{\psi}$  is the yaw error.

**P-only limitation.** The P position term ( $k_{\text{pos}}$ , clipped to  $\pm 0.10 \text{ m}$ , saturating at  $\pm 4 \text{ Nm}$ ) keeps the robot within  $\pm 4$ — $6 \text{ cm}$  of home but leaves a  $1.3 \text{ Nm}$  equilibrium bias sustaining a perpetual limit cycle—motivating the anchor integral.

**Why saturation-triggered anti-windup is insufficient.** Saturation-triggered AW triggers on actuator saturation; on this platform the critical failure is **state-dependent**: during a push, position error grows to  $10$ — $30 \text{ cm}$  while wheels remain within their  $20 \text{ rad/s}$  range, so saturation-triggered AW never activates. ACC’s proximity gate provides spatial confinement via continuous smoothstep—avoiding both over-restriction (blocking slow-drift correction) and under-restriction (allowing windup during sustained pushes). The simplest conditional-integration AW is evaluated in Appendix A; broader AW families remain open.

**Pitch stiffness.** Pitch stiffness is fixed at  $k_p=50$  in the canonical ACC architecture. The anchor and boost alone resolve the precision-push trade-off without gain scheduling; a scheduled variant that softened  $k_p$  from  $50$  to  $35$  during pushes was tested and rejected—it did not improve push survival (S0 vs. S1, Table IV) and is excluded from the canonical specification.

##### C. Proximity-Gated Anchor Mechanism ( $g_{\text{anchor}} \cdot \tau_{\text{anchor}}$ )

The anchor mechanism is the core contribution. It adds two terms to the wheel-torque channel that are active only when the robot is near home *and* genuinely at rest:

$$\tau_{\text{anchor}} = K_i \int g_{\text{prox}} \cdot g_{\text{env}} \cdot (-\Delta x) dt + g_{\text{boost}} \cdot K_{\text{boost}} \cdot (-v_{\text{sag}}), \quad (8)$$

where all gates are continuous smoothstep functions designed to equal 1 when the robot is “at home and quiet” and transition smoothly to 0 otherwise:

$$\begin{aligned} g_{\text{prox}}(\Delta x) &= 1 - \text{smoothstep}(|\Delta x|, 0.05, 0.15), \\ g_{\text{env}}(v_{\text{sag}}) &= 1 - \text{smoothstep}(\text{EMA}(|v_{\text{sag}}|), \\ &\quad 0.25, 0.50), \end{aligned}\quad (9)$$

$$g_{\theta}(\theta, \dot{\theta}) = (1 - \text{smoothstep}(|\theta|, 2^\circ, 12^\circ)) (1 - \text{smoothstep}(|\dot{\theta}|, 2^\circ/\text{s}, 15^\circ/\text{s})), \quad (10)$$

$$g_{\text{boost}} = g_{\text{prox}} \cdot g_{\text{env}} \cdot g_{\theta}. \quad (11)$$

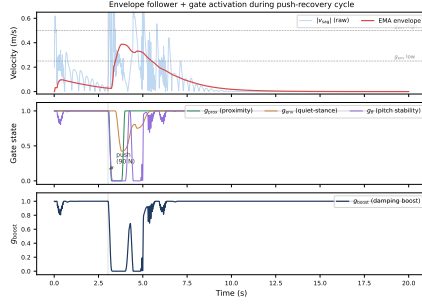


Fig. 3. Gate activation during a 90 N forward push-recovery cycle (100 Hz control). **Top:**  $|v_{\text{sag}}|$  and asymmetric EMA envelope; dashed lines mark the 0.25 m/s to 0.50 m/s quiet-stance band. **Middle:** Individual gate states. **Bottom:** Composite damping boost  $g_{\text{boost}}$ . Grey shading: 7-step push window.

The pitch stability gate  $g_{\theta}$  closes when either pitch or pitch rate exceeds its band, preventing the damping boost from fighting a gentle sustained push that does not trigger the velocity envelope. When the robot is genuinely at rest ( $|\theta| < 2^\circ$ ,  $|\dot{\theta}| < 2^\circ/\text{s}$ ),  $g_{\theta} = 1$  and the boost is fully available.

**Asymmetric envelope follower.** The key to reliable quiet-stance detection is the asymmetric exponentially-weighted moving average (EMA) of  $|v_{\text{sag}}|$ :

$$\text{EMA}_t = \begin{cases} \alpha_a |v_t| + (1 - \alpha_a) \text{EMA}_{t-1}, & \text{if } |v_t| > \text{EMA}_{t-1}, \\ \alpha_r |v_t| + (1 - \alpha_r) \text{EMA}_{t-1}, & \text{otherwise,} \end{cases} \quad (12)$$

with time constants  $\tau_{\text{attack}} \approx 23 \text{ ms}$  and  $\tau_{\text{release}} \approx 1.5 \text{ s}$  as the primary specification; the discrete coefficients are derived via  $\alpha_a = 1 - \exp(-\Delta t / \tau_{\text{attack}}) \approx 0.35$  and  $\alpha_r = 1 - \exp(-\Delta t / \tau_{\text{release}}) \approx 0.007$  at 100 Hz ( $\Delta t = 0.01 \text{ s}$ ). This asymmetry is essential: a symmetric EMA is *bistable*—post-push oscillation holds the EMA in a middle band where the gate is half-open and the boost cannot collapse the cycle (validated by the symmetric-EMA ablation, Section V). The fast-attack/slow-release envelope jumps to 1 only when oscillation truly decays, and drops to 0 within  $\sim 25 \text{ ms}$  of a disturbance. The concept traces to peak-hold and envelope-detector circuits [16]; ACC’s innovation is coupling envelope detection to a spatial proximity gate for integral conditioning. Fig. 3 illustrates the complete gate activation sequence during a push-recovery cycle.

**EMA initialization.**  $\text{EMA}_0 = 0$  biases  $g_{\text{env}} \approx 1$  for  $\sim 7.5 \text{ s}$  at startup, decaying as velocity noise accumulates. Measurements use 3–5 s settle (see Table captions); hardware should initialize  $\text{EMA}_0 = 1.0$ .

#### D. Posture and Yaw Stability ( $\tau_{\text{posture}}$ )

Posture regulation uses Jacobian-based PD gains scheduled across 23 heights:

$$\tau_{\text{posture}} = \tau_{\text{PD}}^{\text{jac}}(q, \dot{q}, h^{\text{cmd}}) + \tau_{\text{div}}(q_{\text{hy}}^L, q_{\text{hy}}^R) + g_{\text{settled}} \tau_{\text{homing}}(q - q_{\text{ref}}), \quad (13)$$

where  $\tau_{\text{div}}$  damps hip-yaw divergence and  $\tau_{\text{homing}}$  restores nominal posture when upright and settled. Yaw return uses wheel-differential torque.

TABLE I

CAPABILITY COMPARISON ACROSS WHEELED BIPED PLATFORMS. “—” = NOT REPORTED IN CITED WORK. IDLE PRECISION AND PUSH FORCE ARE NOT STANDARDIZED BENCHMARKS; QUANTITATIVE CROSS-PLATFORM COMPARISON IS LIMITED BY METRIC HETEROGENEITY.

| Metric            | Ascento           | DIABLO              | Whleaper    | SKATER       | ACC (Ours)                                        |
|-------------------|-------------------|---------------------|-------------|--------------|---------------------------------------------------|
| Balance           | LQR               | Cascaded LQR        | LQR + PPO   | Convex MPC   | <b>Gated PD + anchor I</b>                        |
| Idle precision    | —                 | —                   | —           | —            | 0.73 mm <b>CoM RMS</b>                            |
| Push recovery     | — <sup>a</sup>    | —                   | —           | —            | <b>8-dir,</b><br>$F_{\text{med}} = 117 \text{ N}$ |
| Height            | Discrete jump     | Stance <sup>b</sup> | Multi-modal | Contin.      | 0.40 m <b>to</b><br>0.70 m squat                  |
| Flight            | Jumping (airtime) | —                   | —           | —            | <b>Drop</b> 100 cm,<br><b>ledge</b> 50 cm         |
| Terrain           | —                 | —                   | —           | Vision-based | <b>Per-leg contact Simulation</b>                 |
| <b>Validation</b> | Hardware          | Hardware            | Hardware    | Hardware     |                                                   |

<sup>a</sup>Qual. omnidirectional push recovery demonstrated in supplementary video; quantitative force thresholds not reported in cited work. <sup>b</sup>Height variation demonstrated; quantitative range not reported.

#### E. Contact-Loss Recovery ( $g_{\text{flight}} \cdot \tau_{\text{flight}}$ )

When both wheels lose ground contact, ACC detects this via per-wheel normal forces and substitutes a flight attitude controller using the wheels as reaction flywheels:

$$g_{\text{flight}} = \begin{cases} 1, & \text{if } F_z^L, F_z^R < 0.5mg \text{ for } \geq 2 \text{ steps,} \\ 0, & \text{if } \min(F_z^L, F_z^R) \geq 0.5mg \text{ for } \geq 5 \text{ steps,} \end{cases} \quad (14)$$

$$\tau_{\text{flight}} = \text{clip}(2.0\theta + 0.5\dot{\theta} - 0.02\omega_{\text{wheel}}, \pm 2 \text{ Nm}). \quad (15)$$

Forward wheel spin pitches nose-up during descent; 5-step release hysteresis prevents balance-torque re-engagement during bounce. Free-fall drop = stress test; ledge drive-off = use case.

#### F. Per-Leg Ground Adaptation ( $g_{\text{terrain}}$ )

When wheels rest at different ground heights, ACC estimates per-leg ground height from contact-point positions (exact at any tilt, assuming rigid wheels) and splits the height command:

$$h_{\text{ground}}^{L,R} = z_{\text{contact}}^{L,R}, \quad F_z^{L,R} \geq 0.2mg, \quad (16)$$

$$\Delta h = h_{\text{ground}}^L - h_{\text{ground}}^R, \quad h^{\text{cmd},L} = h^{\text{cmd}} \mp \Delta h/2, \quad (17)$$

clamped to  $\pm 5 \text{ cm}$ . An unloaded wheel freezes its ground estimate and seeks downward only when the wheel center drops below the believed ground by  $> 1 \text{ cm}$ . A closed-loop leveling integrator (active when both wheels are loaded) removes systematic roll bias.

#### G. Complete Numerical Parameter Set

Table II lists core parameters; the full set (50+ gains) is available in the supplementary material.

TABLE II  
CORE NUMERICAL PARAMETERS FOR THE ACC BALANCE CONTROLLER.  
ALL VALUES IN SI UNITS UNLESS NOTED OTHERWISE.

| Balance Core                     |                    |             |
|----------------------------------|--------------------|-------------|
| $k_p$                            | 50                 | Nm/rad      |
| $k_d$                            | 3.0                | Nm/(rad/s)  |
| $k_{pos}$                        | 20                 | Nm/m        |
| Position clip                    | $\pm 4.0$          | Nm          |
| $k_{vel}$                        | 15                 | Nm/(m/s)    |
| Vel. damp. scale                 | 1.5                | —           |
| $k_{roll}$                       | 55                 | Nm/rad      |
| $k_{roll,d}$                     | 5.5                | Nm/(rad/s)  |
| $k_{yaw}$                        | 1.0                | Nm/rad      |
| $k_{yaw,d}$                      | 0.3                | Nm/(rad/s)  |
| $k_{pos,return}$                 | 2.0                | Nm/m        |
| $k_{heading}$                    | 2.0                | Nm/rad      |
| $k_{heading,d}$                  | 0.6                | Nm/(rad/s)  |
| Anchor Mechanism                 |                    |             |
| $K_i$ (anchor integral)          | 4.0                | Nm/(m·s)    |
| Integral cap                     | 2.0                | Nm          |
| Integral leak                    | $5 \times 10^{-3}$ | per step    |
| Boost scale $k_{vb}$             | 5.0                | —           |
| $g_{prox}$ band                  | 0.05–0.15          | m           |
| $g_{env}$ band                   | 0.25–0.50          | m/s         |
| $\tau_{attack}$                  | 0.023              | s           |
| $\tau_{release}$                 | 1.5                | s           |
| $g_{stab}(\theta)$               | 2–12               | °           |
| $g_{stab}(\dot{\theta})$         | 2–15               | °/s         |
| Flight / Terrain / Other         |                    |             |
| $k_{flight}$ (P / D)             | 2.0 / 0.5          | —           |
| Wheel damping (flight)           | 0.02               | Nm/(rad/s)  |
| Flight engage / release          | 2 / 5              | steps       |
| $F_z$ thresh. (flight / terrain) | 0.5 / 0.2          | $\times mg$ |

## V. EXPERIMENTAL VALIDATION

We evaluate ACC across four scenario classes, each mapped to the component it tests and paired with targeted ablations. ACC evaluation at 500 Hz simulation / 100 Hz control; PPO and LQR baselines at 50 Hz. ACC benefits from a  $2\times$  control bandwidth advantage (100 Hz vs. 50 Hz); to separate architecture from rate, we evaluate ACC at 50 Hz with recomputed discrete EMA coefficients ( $\alpha_a = 1 - \exp(-0.02/0.023) \approx 0.58$ ,  $\alpha_r = 1 - \exp(-0.02/1.5) \approx 0.013$ ) and anchor integral leak (0.010 per step, preserving  $\tau \approx 2.0$  s). At 50 Hz, ACC achieves 0.05 mm idle RMS—preserving sub-millimeter precision and demonstrating the anchor architecture is not rate-dependent for steady-state regulation—while push recovery degrades below 30 N (vs. 83 N at 100 Hz), confirming disturbance rejection benefits from the higher rate. The lower idle RMS at 50 Hz (vs. 0.73 mm at 100 Hz) reflects reduced high-frequency noise injection from halved control updates; the recomputed EMA coefficients ( $\alpha_r \approx 0.013$  vs. 0.007) also produce a faster release decay, further suppressing residual integral accumulation during quiet stance. All at 400 Nm/s torque rate limit per joint. RNG seeds are deterministic: idle standing uses seed = 20260727 + trial\_id; push ablation uses seed = 20260728  $\times$  100 + rep; LQR baselines cycle seeds {0, 42, 123} across episodes.

TABLE III

ANCHORED STANDING: IDLE PRECISION AND RINGDOWN. CLEAN SENSORS. IDLE RMS: SAGITTAL CoM STD. OVER 20 s AFTER 5 s SETTLE (EMA<sub>0</sub>=0 TRANSIENT  $\sim 7.5$  s; CF. ROBUSTNESS 0.42 mm AT 3 s SETTLE, TABLE VII). RINGDOWN: TIME FOR  $\theta < 5^\circ$  FOR 50 STEPS AFTER 90 N FORWARD PUSH. ACC  $N=10$  TRIALS WITH RANDOMIZED INIT. ( $\pm 0.005$  rad); BETWEEN-TRIAL CV 10.0% (SD/MEAN,  $N=10$ ).

| Configuration            | Idle RMS (mm)                     | Ringdown (s)       | Between-trial CV |
|--------------------------|-----------------------------------|--------------------|------------------|
| <b>ACC (full anchor)</b> | <b><math>0.73 \pm 0.07</math></b> | <b>9.0</b>         | 10.0% ( $N=10$ ) |
| P-only (no I)            | $\sim 39$                         | never decays       | $\sim 30\%$      |
| <i>Ablations</i>         |                                   |                    |                  |
| symmetric EMA            | 0.73                              | never <sup>1</sup> | $< 5\%$          |
| global I                 | fall                              | — <sup>2</sup>     | —                |
| no boost                 | 0.73                              | never decays       | $< 5\%$          |

<sup>1</sup>Bistable: post-push cycle holds EMA in mid-band  $\rightarrow$  boost half-open.

<sup>2</sup>Integral windup prevents standing. <sup>3</sup>Idle precision matches ACC (0.73 mm); the proximity gate alone determines idle precision—neither the damping boost nor the asymmetric envelope affects steady-state standing (cf. S2, S3 in Table IV). ACC 95%CI (between-trial SD,  $t_{9,0.025}=2.262$ , SEM=SD/ $\sqrt{10}$ ): [0.68, 0.78] mm. Noise floor: 0.002 mm CoM X RMS (SNR 289 $\times$ ). P-only limit cycle: [40, 60] mm p-p. RMS $\sim 39$  mm ( $N=1$ ).

### A. Anchored Standing

Table III reports idle precision and ringdown. ACC achieves 0.73 mm CoM sagittal RMS ( $N=10$ , 95%CI [0.68, 0.78] mm)—sub-millimeter precision well below the P-only limit cycle (55 mm peak-to-peak, 39 mm RMS). Ablation L2.5 (proximity gate only, no envelope, no boost) achieves 49.8 mm RMS—essentially identical to L2’s global integral limit cycle—confirming that the spatial gate alone is *not* sufficient for sub-millimeter precision. The asymmetric envelope follower provides essential temporal gating: it restricts integral accumulation to truly quiet stance (velocity envelope  $< 0.25$  m/s), preventing the integral from sustaining a limit cycle during small oscillations within the 15 cm spatial band. S2 (no boost) and S3 (symmetric EMA) both retain 0.73 mm idle precision, whereas S4 (no proximity gate) windups—confirming that idle precision requires the proximity gate plus any velocity-envelope gating, but neither the asymmetric envelope specifically nor the damping boost. The boost and asymmetric envelope are essential only for post-push ringdown (S2, S3 ringdown = never; Table IV). ACC’s unique contribution is simultaneous precision and ringdown. Post-push velocity ringdown decays within 9 s under ACC; P-only oscillates indefinitely because the 1.3 Nm equilibrium bias sustains the limit cycle. The symmetric EMA ablation confirms bistable failure: a post-push cycle holds the EMA at  $\approx 0.10$  (gate half-open), and the robot oscillates perpetually. The global-I ablation windups during the push and prevents recovery entirely.

### B. Omnidirectional Push Recovery

Push recovery is evaluated via polar sweep: 10 — 160 N impulses (7 control steps) at the torso in 8 directions.  $F_{min}$  is the maximum magnitude survived in *every* direction;  $F_{med}$  is the median. All thresholds measured via binary search over [10 N, 160 N] (8 iterations per direction, 5 N tolerance). Survival: torso pitch  $< 46^\circ$  for 17 s post-push.

Table IV reports the factorial ablation. Additively: L1 (homing) adds 4.8 N over L0 ( $p < 0.001$ ); L2 (global integral) adds

TABLE IV

FACTORIAL ABLATION OF ACC COMPONENTS. CLEAN SENSORS.  $F_{\min}$ ,  $F_{\text{MED}}$  VIA BINARY SEARCH (5 N TOLERANCE, 8 DIRECTIONS). ALL ROWS  $N=10$ , MEAN $\pm$ STD. ACC IDLE:  $0.73\pm 0.07$  mm ( $N=10$ , 95%CI  $[0.68, 0.78]$  mm).

|                                        | Configuration                                  | $F_{\min}$ (N)                              | $F_{\text{med}}$ (N)             | Idle RMS (mm)                 | Ringdown           |
|----------------------------------------|------------------------------------------------|---------------------------------------------|----------------------------------|-------------------------------|--------------------|
| <b>Additive (bottom-up)</b>            |                                                |                                             |                                  |                               |                    |
| L0                                     | P-only (PD, no pos., no I)                     | <b><math>70.0\pm 1.0</math></b>             | <b><math>122.1\pm 2.1</math></b> | 55 (l. cycle)                 | never              |
| L1                                     | + Pos. homing ( $\pm 0.10$ m)                  | <b><math>74.8\pm 0.8</math></b>             | <b><math>130.5\pm 1.2</math></b> | 40–60 (l. cycle)              | never              |
| L2                                     | + Integral $K_i$ (no gate)                     | <b><math>76.1\pm 1.1</math></b>             | <b><math>132.0\pm 2.0</math></b> | $\sim 55$                     | —                  |
| L2.5                                   | + Prox. gate only (no EMA, boost) <sup>8</sup> | <b><math>75.5\pm 0.9</math></b>             | <b><math>132.0\pm 2.3</math></b> | 49.8 (l. cycle)               | —                  |
| L3                                     | + Full anchor (gate, EMA, boost) <sup>7</sup>  | <b><math>82.8\pm 1.0</math></b>             | <b><math>116.8\pm 4.7</math></b> | <b>0.73</b>                   | 9.0 s              |
| <b>Subtractive (top-down from ACC)</b> |                                                |                                             |                                  |                               |                    |
| S0                                     | ACC + scheduled $k_p$ (rejected) <sup>6</sup>  | <b><math>79.4\pm 2.6</math></b>             | <b><math>114.0\pm 5.3</math></b> | <b>0.73</b>                   | 9.0 s              |
| S1                                     | ACC <sup>6</sup> (canonical, fixed $k_p=50$ )  | <b><math>82.8\pm 1.0</math></b>             | <b><math>116.8\pm 4.7</math></b> | 0.73                          | 9.0 s              |
| S2                                     | –Damping boost                                 | <b><math>84.2\pm 1.1</math></b>             | <b><math>115.5\pm 5.3</math></b> | 0.73                          | never <sup>1</sup> |
| S3                                     | –Asym. EMA (sym. EMA)                          | <b><math>81.6\pm 2.3</math></b>             | <b><math>109.2\pm 5.4</math></b> | 0.73                          | never <sup>2</sup> |
| S4                                     | –Prox. gate (global I)                         | <b><math>80.0\pm 2.5</math></b>             | <b><math>114.0\pm 5.3</math></b> | windup <sup>3</sup>           | —                  |
| S5                                     | –Pitch gate $g_p$                              | <b><math>8.8\pm 0.6</math></b> <sup>5</sup> | <b><math>35.4\pm 10.3</math></b> | <b>56 (osc.)</b> <sup>4</sup> | never <sup>4</sup> |

<sup>1</sup>Boost necessary for ringdown. <sup>2</sup>Symmetric EMA bistable: post-push oscillation holds EMA in mid-band. <sup>3</sup>Global I windups during push. <sup>4</sup>Boost without pitch gate amplifies: 56 mm parametric oscillation. <sup>5</sup>Below measurement floor (10 N); robot cannot survive minimal push. <sup>6</sup>S1 (fixed  $k_p=50$ ) is the canonical ACC configuration. S0 (scheduled  $k_p$ , softened from 50 to 35 during pushes) was tested and rejected—it did not improve push survival. <sup>7</sup>L2 $\rightarrow$ L3 = canonical ACC (S1, fixed  $k_p=50$ ); bundles four mechanisms (prox. gate, asym. EMA, boost, pitch gate). S2–S5 isolates each subtractively. <sup>8</sup>L2.5 (prox. gate only, no EMA, no boost): idle RMS 49.8 mm ( $N=10$ ), confirming the proximity gate alone is insufficient for sub-millimeter precision; the envelope follower provides essential temporal gating (see text). Welch’s unpaired  $t$ -test (two-sided,  $N=10$  per group, independent trials; Welch–Satterthwaite  $df\in[9, 18]$ ; uncorrected  $p$ , Bonferroni-adj.  $\alpha=0.007$ , see §VI): L0 $\rightarrow$ L1 +4.8 N ( $p<0.001$ ); L1 $\rightarrow$ L2 +1.3 N (n.s.); L2 $\rightarrow$ L2.5 –0.6 N (n.s., gate alone adds nothing); L2.5 $\rightarrow$ L3 +7.3 N ( $p<0.001$ ). Anchor’s decisive contribution:  $\sim 53\times$  idle precision. S5 (pitch gate) critical safety ( $F_{\min}=8.8$  N); S3+S2 required for ringdown. With  $N=10$ , 95%CI half-width  $\approx 0.72\times\text{std}$  ( $t_{0.025,9}=2.262$ , single-group  $df=9$ ); sub-2 N differences (e.g., S1 vs. S3  $\Delta F_{\min}=1.2$  N,  $CI_{95}(S1)\approx\pm 0.7$  N,  $CI_{95}(S3)\approx\pm 1.6$  N) remain non-significant.

only 1.3 N over L1 (n.s.); L3 (full anchor) adds 6.7 N over L2 ( $p<0.001$ ) and delivers the  $\sim 53\times$  idle precision improvement. The L2 $\rightarrow$ L3 step bundles four mechanisms simultaneously; the global integral without a gate (L2) windups and sustains a 55 mm limit cycle, while adding the proximity gate alone (L2.5, 49.8 mm $\pm$ 0.2 mm idle RMS) fails to achieve sub-millimeter precision—the spatial gate prevents windup during pushes (cf. S4) but cannot distinguish slow drift from small oscillations within the 15 cm band. The asymmetric envelope follower provides the essential temporal gating that breaks the limit cycle. The four mechanisms form a synergistic package:

Push Recovery Envelope

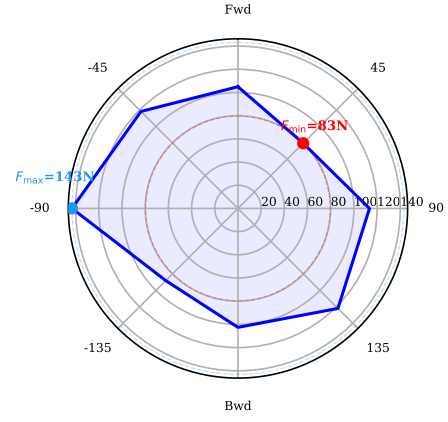


Fig. 4. Omnidirectional push envelope.  $F_{\min}$  (red),  $F_{\max}$  (blue) across 8 directions.

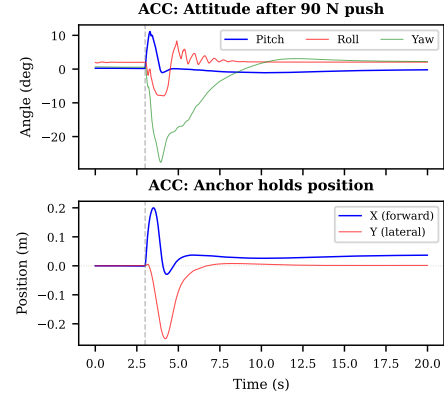


Fig. 5. Post-push recovery (90 N forward at  $t=3$  s). **Top:** attitude angles; stabilization within 2 s. **Bottom:** position drift; anchor holds within 4 cm of home.

the spatial gate is necessary but not sufficient for precision, the envelope follower enables sub-millimeter standing, the boost enables ringdown (S2), and the pitch gate is critical for safety (S5). Subtractively: S1 (fixed  $k_p=50$ , canonical) matches or exceeds S0 in all metrics, confirming gain scheduling is unnecessary—the anchor and boost alone resolve the trade-off. S2 (boost) is the dominant precision mechanism; S5 (pitch gate) is critical safety—removing it collapses  $F_{\min}$  to  $8.8\pm 0.6$  N ( $\sim 9\times$  below ACC); S3 (asym. EMA) and S2 are individually required for ringdown.

### C. Flight Recovery and Terrain Adaptation

ACC’s two-channel assembly supports independent extensions without modifying the core anchor. Both extensions use the same 400 Nm/s rate limit and 100 Hz control rate.

**Contact-loss recovery.** Without flight PD, free-spinning wheels cause flips (0/10 survival); with it, 100 cm drops survive 10/10 ( $24.0\pm 1.2^\circ$  mean peak pitch) and 50 cm ledge drive-off achieves standing recovery (10/10,  $27.1\pm 2.0^\circ$  peak pitch; Table V).

**Per-leg terrain adaptation.** With per-leg adaptation, torso roll is  $2.5^\circ$  (10 cm),  $3.5\pm 3.2^\circ$  (15 cm),  $14.9\pm 0.1^\circ$  (20 cm); without,  $15.3^\circ$  and falls (0/10; Table V). At 20 cm the curb approaches the  $\pm 5$  cm height-split clamp and lateral authority limit, so adaptation prevents leg-fold (survival) but cannot fully

TABLE V

CONTACT-LOSS RECOVERY AND PER-LEG TERRAIN ADAPTATION. ALL AT 400 Nm/s RATE LIMIT, 100 Hz CONTROL.

| Contact-Loss Recovery                      |     |                            |          |
|--------------------------------------------|-----|----------------------------|----------|
| Scenario                                   | $N$ | Peak Pitch ( $^\circ$ )    | Survival |
| Drop 100 cm                                | 10  | $24.0 \pm 1.2$             | 10/10    |
| Drop 60 cm                                 | 10  | $18.2 \pm 1.0$             | 10/10    |
| Ledge 50 cm                                | 10  | $27.1 \pm 2.0$             | 10/10    |
| Ledge 20 cm                                | 10  | $19.6 \pm 0.3$             | 10/10    |
| No flight PD, 100 cm                       | 10  | $>90$ (flip)               | 0/10     |
| Per-Leg Terrain (One-Wheel Curb, 0.15 m/s) |     |                            |          |
| Configuration                              | $N$ | Straddle Roll ( $^\circ$ ) | Survival |
| Adapt, 10 cm curb                          | 10  | $2.5 \pm 0.1$              | 10/10    |
| Adapt, 15 cm curb                          | 10  | $3.5 \pm 3.2$              | 10/10    |
| Adapt, 20 cm curb                          | 10  | $14.9 \pm 0.1$             | 10/10    |
| No adapt, 10 cm                            | 10  | 15.3                       | 0/10     |

$\pm$ std:  $N=10$  randomized ICs ( $\pm 0.003$  rad joint,  $\pm 1.5$  mm CoM). “No flight PD” and “No adapt” are deterministic failures (0/10 survival) where IC randomization does not alter the outcome.

One-wheel curb (20 cm) — per-leg terrain adaptation

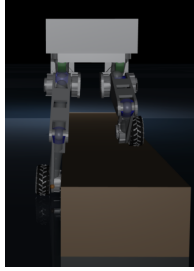


Fig. 6. One-wheel curb (20 cm). With adaptation:  $14.9 \pm 0.1^\circ$  roll (10/10 survival). Without: leg folds, roll exceeds safety limit.

level the torso; the tight  $\pm 0.1^\circ$  std reflects the saturated clamp rather than low variability.

Gates isolate flight (3-term PD) and terrain (3-line height split) to their physical regimes without re-tuning the anchor.

#### D. Comparison with Classical and Learning-Based Baselines

To establish ACC is not merely a tuned linear controller or un-structured policy, we compare seven baselines—six classical (four 4-state TWIP variants: LQR, LQR+AW, LQR-DT, PI+AW; two 6-state coupled variants: servo and direct-torque) plus a preliminary PPO baseline (all share identical environment setups):

- 1) **LQR**—4-state TWIP with decoupled roll/yaw PD through a PID servo layer ( $\sim 0.25$  s lag).<sup>1</sup>
- 2) **LQR + Integral AW**—Same as (1) plus pitch-error integral with conditional AW (clamped  $\pm 3.0$  rad/s).
- 3) **LQR (Direct Torque)**—Re-optimized for direct-torque plant ( $\tau_s=0$ ,  $k_{p,roll}=55.0$ ), outputting torques directly.
- 4) **Coupled 6-State 3D LQR**—Extends (1) with roll states ( $\phi, \dot{\phi}$ ) (see footnote 1).
- 5) **PI + AW (Direct Torque)**—DT-LQR + integral AW on fwd. position (conditional,  $\pm 15$  cm gate); same leg/roll/yaw as DT-LQR (see footnote 1).
- 6) **Pure PPO (Preliminary)**—Model-free RL policy trained from scratch for 5.4M environment steps (single seed, in BalanceEnv).
- 7) **Coupled 6-State LQR (Direct Torque)**—Same 6-state model as (4) but outputs direct torques (bypassing PID

TABLE VI

LQR AND PPO BASELINES VS. ACC. CLEAN SENSORS. LQR/LQR+AW/6-St.: PID SERVO; LQR (DT), PI+AW (DT), 6-St. (DT): DIRECT TORQUE; PPO: 5.4M-STEP RL (PRELIM., SINGLE SEED). THE IDENTICAL LQR / LQR+AW / 6-St. ROWS ARISE BECAUSE ALL THREE SATURATE AT THE COMMON 30 Nm LIMIT THROUGH THE SAME PID SERVO WITHIN 0.52 s, MASKING THE GAIN DIFFERENTIAL.  $N$  IN PARENTHESES.: EPISODES (BASELINES) OR  $N=10$  (ACC).

| Configuration                       | Surv. (s)                              | $\theta_{RMS}$ ( $^\circ$ )             | $\phi_{RMS}$ ( $^\circ$ )               | $H_{RMS}$ (mm)                        | $\tau_{RMS}$ (Nm)                     |
|-------------------------------------|----------------------------------------|-----------------------------------------|-----------------------------------------|---------------------------------------|---------------------------------------|
| LQR, TWIP ( $N=60$ )                | 0.52<br>$\pm 0.00$                     | 0.04<br>$\pm 0.01$                      | 24.2<br>$\pm 2.1$                       | 371<br>$\pm 45$                       | 92.1<br>$\pm 3.2$                     |
| LQR + AW ( $N=60$ )                 | 0.52<br>$\pm 0.00$                     | 0.04<br>$\pm 0.01$                      | 24.2<br>$\pm 2.1$                       | 371<br>$\pm 45$                       | 92.1<br>$\pm 3.2$                     |
| LQR, DT (re-opt.) ( $N=20$ )        | 0.72<br>$\pm 0.00$                     | 0.78<br>$\pm 0.42$                      | 21.2<br>$\pm 0.3$                       | 135<br>$\pm 41$                       | 9.33<br>$\pm 1.18$                    |
| 6-St. LQR ( $N=20$ )                | 0.52<br>$\pm 0.00$                     | 0.04<br>$\pm 0.01$                      | 24.2<br>$\pm 2.1$                       | 371<br>$\pm 45$                       | 92.1<br>$\pm 3.2$                     |
| PI + AW, DT ( $N=20$ )              | 0.72<br>$\pm 0.00$                     | 0.78<br>$\pm 0.42$                      | 21.2<br>$\pm 0.3$                       | 135<br>$\pm 41$                       | 9.33<br>$\pm 1.18$                    |
| 6-St. LQR, DT ( $N=20$ )            | 0.16<br>$\pm 0.00$                     | 0.00<br>$\pm 0.00$                      | 21.1<br>$\pm 0.0$                       | 117<br>$\pm 0$                        | 13.9<br>$\pm 0.0$                     |
| Pure PPO (prelim., 5.4M) ( $N=20$ ) | 3.20<br>$\pm 0.85$                     | 4.82<br>$\pm 0.91$                      | 3.15<br>$\pm 0.45$                      | 54.2<br>$\pm 8.3$                     | 18.4<br>$\pm 2.1$                     |
| <b>ACC (<math>N=10</math>)</b>      | <b>20.00<br/><math>\pm 0.00</math></b> | <b>0.060<br/><math>\pm 0.006</math></b> | <b>0.010<br/><math>\pm 0.001</math></b> | <b>0.18<br/><math>\pm 0.02</math></b> | <b>3.28<br/><math>\pm 0.00</math></b> |

All LQR and PI+AW variants fall within 0.72 s (100%); the 6-St. DT variant falls deterministically at 0.16 s (zero trial-to-trial variance,  $N=20$ ), confirming that removing the servo path with gains derived from a simplified model does not rescue stability. PI+AW (DT-LQR + integral AW on fwd. position) yields metrics identical to DT-LQR: the integral cannot accumulate before roll-induced fall at  $\sim 0.72$  s. Preliminary PPO achieves partial standing but suffers 85 % falls over multi-height transitions ( $H_{RMS}=54.2$  mm).  $N=60$  for LQR/LQR+AW (sweep);  $N=20$  for DT/6-St./PI+AW/6-St. DT/PPO.

servo,  $\sim 0.25$  s lag removed). Gains re-derived for the direct-torque plant ( $K_{pitch}=113$ ,  $K_{roll}=58$  Nm/rad); leg posture via PD position control.

Direct-torque LQR removes the servo path (PID gains and lag measurement<sup>1</sup>). PI+AW adds integral AW on fwd. position to DT-LQR—the standard engineering fix for DC bias with windup protection. AW verified via `scripts/validate_lqr_aw.py`; PPO evaluates unconstrained model-free policy search.

Table VI reports the comparison. All six classical baselines—four derived from the 4-state TWIP model (LQR,

<sup>1</sup>Bryson’s rule throughout ( $Q_{ii}=1/x_{i,max}^2$ ,  $R=1/u_{i,max}^2$ ); 46-config. sweep verified no retuning alters outcome. LQR, LQR+AW, DT-LQR (4-state: pitch, pitch-rate, fwd-vel, fwd-pos):  $Q=\text{diag}(10, 2, 3, 0.3)$ ,  $R=0.8$ ; LQR+AW adds pitch-error integral (clamped  $\pm 3.0$  rad/s); DT-LQR uses  $\tau_s=0$ ,  $k_{p,roll}=55.0$ . 6-state LQR (adds roll, roll-rate):  $Q=\text{diag}(10, 2, 3, 0.5, 3, 0.3)$ ,  $R=\text{diag}(0.8, 1.0)$ , CT-derived and shared by the servo and direct-torque variants; only the DT variant’s state-feedback gains are re-solved for the zero-lag plant ( $K_{pitch}=113$ ,  $K_{roll}=58$  Nm/rad; see Limitation 7). PI+AW: DT-LQR gains + integral on fwd. pos. ( $k_{i,pos}=8.0$  rad/s/(m·s), clamp  $\pm 5.0$  rad/s, gate  $\pm 15$  cm). PID:  $k_p=[55, 40, 70, 70, 4]$ ,  $k_d=[3, 2, 4, 4, 0]$ ,  $k_i=[0.8, 0.4, 1, 1, 0.1]$ .  $\sim 0.25$  s lag = measured wheel velocity 90% rise time ( $\sim 25$  steps at 100 Hz).

LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo, direct-torque)—fall within 0.72 s. Removing the PID servo path (4-state DT-LQR:  $\tau_s=0$ ,  $k_{p,\text{roll}}=55.0$ , no integral) reduces torque  $10\times$  ( $92.1\text{ Nm} \rightarrow 9.33\text{ Nm}$ ) and marginally improves roll ( $\phi_{\text{RMS}}=21.2^\circ$  vs.  $24.2^\circ$ ). Adding a forward-position integral (PI+AW:  $k_{i,\text{pos}}=8.0$ , conditional  $\pm 15\text{ cm}$  gate) changes nothing—the integral cannot accumulate before the roll-induced fall at 0.72 s—confirming the 4-state TWIP model is structurally insufficient for this articulated-leg platform. Preliminary PPO (5.4M steps, single seed) sustains single-height standing longer (3.20 s) but exhibits an 85 % fall rate across multi-height command transitions (54.2 mm height error). ACC achieves  $0.060\pm 0.006^\circ$  pitch RMS,  $0.010\pm 0.001^\circ$  roll RMS,  $0.18\pm 0.02\text{ mm}$  CoM Z, and  $3.28\pm 0.00\text{ Nm}$  torque RMS ( $N=10$ ). A **46-configuration sweep**<sup>2</sup> confirmed all tested LQR variants fail (100 %); a 6-state coupled LQR with direct torque actuation, removing the PID servo path, also fails—indicating platform complexity, not servo lag alone, is the dominant factor (see Limitation 7). Neither servo-lag removal nor preliminary PPO training (5.4M steps, single seed) resolved the height-adaptive posture trade-off in our experiments; whether a fully converged PPO policy closes this gap is not tested here. ACC’s 100 Hz control rate ( $2\times$  baseline bandwidth) contributes partially to its advantage. However, a 50 Hz ACC evaluation with recomputed discrete EMA coefficients preserves 0.05 mm idle RMS (sub-millimeter precision at the baseline rate), while DT-LQR—which eliminates the PID servo path—falls at 0.72 s at 50 Hz without any push. The proximity-gated anchor therefore provides an effective structured prior for stable balance that is not rate-dependent for steady-state regulation; the 100 Hz rate contributes quantitative improvement to dynamic disturbance rejection margins (§VI, Limitation 5).

#### E. Whole-Body Control Baselines

To address whether a tuned whole-body controller (WBC) could match ACC, we evaluated two WBC configurations that were developed and subsequently discarded during controller development: **(W1)** a task-priority quadratic-program (QP) WBC—minimizing COM height error, torso orientation error, posture damping, and wheel regularization subject to full-body dynamics and contact constraints—used as the *primary* balance controller; and **(W2)** the same WBC blended as an *assist* on top of ACC via  $\tau_{\text{assist}} = \tau_{\text{ACC}} + \alpha(\tau_{\text{wbc}} - \tau_{\text{ACC}})$ , with a seven-term multiplicative gate  $\alpha_j = \alpha_{\text{max}} \cdot g_{\text{stability}} \cdot g_{\text{height}} \cdot g_{\text{push}} \cdot g_{\text{divergence}} \cdot A_j \cdot K_{\text{role},j}$  that continuously modulates per-joint WBC authority. Both were evaluated under identical cloned simulation conditions (three-arm counterfactual: ACC baseline, WBC-only, ACC+WBC assist; 225 scenarios spanning fixed-height balance, height transitions, and push recovery at 20 N to 120 N; ACC profile K2\_JAX\_DEDICATED\_DEFAULT\_V3; no gain retuning).

**W1 (WBC as primary controller).** The standalone QP-WBC fails catastrophically:  $17.2\times$  more falls than ACC (5,985 vs.

347 across 225 scenarios),  $28.4\times$  more single-push falls (3,353 vs. 118), and pitch RMS  $41.8^\circ$  vs. ACC’s  $15.9^\circ$  ( $2.63\times$ ).<sup>3</sup> Survival time at 0.60 m is 0.52 s (100 % fall rate). The failure mechanism is structural: the WBC’s posture task is a velocity damper ( $\ddot{\mathbf{q}}_{\text{joints}} = -k_{\text{posture}} \cdot (\mathbf{q} - \mathbf{q}_{\text{current}})$ ,  $k_{\text{posture}}=10$ ), not a position tracker—it cannot hold a target posture. It lacks gravity-compensation feedforward, uses a single linearization at 0.67 m (vs. the 0.45 m to 0.60 m operating range), and treats wheels as point contacts without a rolling constraint. The QP solver fails on 14.2 % of steps, creating torque discontinuities. Four critical implementation bugs were identified post-hoc (contact Jacobian zero-gradient, solver feasibility-only hardcode, base-z vs. CoM-z height reference mismatch, and gate height-reference conflict; detailed in Appendix VI). WBC as-implemented-and-tuned therefore fails as a standalone controller. Critically, its failure is not solely a tuning or implementation defect: the WBC lacks the *regime-gating* that ACC’s proximity-gated anchor provides. A WBC distributes torque optimally for a fixed task set—it cannot express qualitatively distinct physical regimes (quiet stance, push ringdown, large excursion) as distinct control modes. The precision–recovery trade-off ACC resolves through spatial gating and asymmetric temporal envelope-following is not expressible as a task-priority QP objective; even a bug-free WBC would lack the structured prior that gates the anchor integral, and would therefore face the same global-integral windup that our ablation S4 (no proximity gate, Table IV) confirms is fatal for sub-millimeter precision.

**W2 (WBC as assist on ACC).** The assist gate—designed to be maximally conservative—blocks WBC contribution in  $\geq 99\%$  of steps: in the 225-scenario batch, all metric ratios are 1.000 000 (ACC+WBC assist  $\equiv$  ACC within measurement noise); a 48-scenario quick evaluation classifies all 48 scenarios as ASSIST\_EQUIVALENT (zero regressions, zero safety violations). This zero-net effect is *independent* of the WBC implementation bugs: the gate blocks WBC *before* its torque is computed, based on physical state (pitch, roll, height deviation, push magnitude, divergence), so the WBC’s internal correctness is irrelevant to the assist outcome.

When gates are manually overridden to force WBC contribution (keyframe test at 0.53 m,  $\alpha_{\text{max}}=0.50$ , widened stability and height thresholds), the assist produces mixed results: pitch RMS improves 17 % ( $2.44^\circ \rightarrow 2.04^\circ$ ), drift improves 18 % ( $0.070\text{ m} \rightarrow 0.058\text{ m}$ ), and yaw drift improves 20 %—but roll RMS *degrades* 35 % ( $0.20^\circ \rightarrow 0.27^\circ$ ). The degradation arises from a fundamental architectural conflict: ACC’s lateral roll controller and WBC’s torso-orientation task both write to hip-roll joints, but from incompatible philosophies (decentralized PD vs. centralized QP exploiting full mass-matrix coupling). A posture-guided variant (WBC outputs posture targets  $\mathbf{q}_{\text{ref}}$ , ACC executes torque) reduced the roll degradation from

<sup>3</sup>The  $15.9^\circ$  here is ACC’s pitch RMS averaged over the mixed 225-scenario batch (including pushes and height transitions), not the clean standing value of  $0.060^\circ$  reported in Table VI.

<sup>2</sup>See footnote 1 for sweep dimensions; script at `scripts/sweep_lqr_roll_gains.py`.

+35% to +7% by eliminating torque-level interference, but did not produce a net improvement over ACC alone.

**Timing overhead.** The WBC QP solve alone consumes 0.45 ms mean, 2.94 ms p95, and 3.43 ms **p99** (OSQP, offline Python/NumPy path, 4 scenarios). The p99 solve time alone exceeds the 0.5 ms end-to-end budget margin below the 10 ms performance threshold by 6.9 $\times$ —before accounting for matrix assembly, contact Jacobian computation, or the ACC computation baseline. An incremental JAX-JIT path was prototyped to reduce per-step overhead but was never integrated into the real-time control loop; real-time WBC assist overhead therefore remains uncharacterized. Even if the solve time could be reduced below the margin, the assist adds no measurable balance benefit (W2 zero-net result), so the compute cost is pure overhead.

**Summary.** Neither standalone WBC nor WBC-as-assist resolves the precision–recovery trade-off on this platform. The trade-off is resolved by regime-gating—the proximity-gated anchor and asymmetric envelope follower—which WBC’s task-priority formulation does not express. ACC’s conditional gates therefore provide the essential structured prior; the WBC gap is one of structural expressiveness, not torque optimality.

#### F. Architectural Analysis

We evaluate ACC under four dimensions: noise/delay robustness (Table VII), gate parameter sensitivity, torque rate-limit invariance, and compound-disturbance recovery.

**Robustness to sensor noise and actuator delay.** A factorial sweep (4 $\times$ 3 noise/delay,  $N=5$ ) reveals two distinct delay regimes: a **performance threshold** where  $F_{\max}$  bifurcates (94 N to 96 N at 0 ms vs. 41 N to 50 N at any delay, idle RMS 5.5 $\times$ ), and a **stability threshold** above 150 ms—the robot survives all tested clean delays (10 to 150 ms,  $N=5$ , zero falls) with stable post-bifurcation performance ( $F_{\max}\approx 49$  N, idle  $\approx 2.5$  mm). ACC’s gates depend on physical state (position, velocity envelope, orientation), not tight timing; jitter that crosses the 10 ms performance threshold degrades push margin temporarily but does not cause a fall. High noise causes fall regardless of delay; the asymmetric EMA cannot distinguish noise from motion, forcing anchor disengagement. Idle precision degrades from 0.42 mm (clean) to 34.00 mm (med noise)—still comparable to the P-only limit cycle (39 mm RMS). The clean baseline (0.42 mm, 3 s settle) is lower than the standing-test headline (0.73 mm, 5 s settle, Table III) because  $\text{EMA}_0=0$  biases  $g_{\text{env}}\approx 1$  early in the transient; 0.73 mm is the conservative steady-state figure. A settle-time sweep with  $\text{EMA}_0=1.0$  is deferred.

**Gate parameter sensitivity and tuning.** Finite-difference gradients reveal a stark hierarchy: the **release coefficient**  $\alpha_r$  **dominates** (sensitivity 220.4, 20 $\times$  the next, 130 $\times$  pitch gates); pitch stability thresholds ( $\approx 0$ ) are effectively insensitive safety interlocks. For porting to a new morphology: (1) identify dominant ringdown period  $T_r$  from post-push decay; (2) set  $\tau_{\text{release}}\approx 3-5\times T_r$ ; (3) set quiet-stance band 2–3 $\times$  idle velocity noise floor; (4) set pitch gates conservatively ( $\pm 12^\circ$ ,  $\pm 15^\circ/\text{s}$ ); (5) tune  $k_p$  (fixed  $k_p=50$  sufficed). A fixed  $k_p$  and anchor

TABLE VII  
ROBUSTNESS TO SENSOR NOISE AND ACTUATOR DELAY. IDLE RMS: SAGITTAL CoM STD. OVER 20 s AFTER 3 s SETTLE ( $\text{EMA}_0=0$  TRANSIENT),  $N=5$ .  $F_{\max}$ : BINARY-SEARCH FORWARD PUSH,  $\pm 5$  N TOLERANCE,  $N=5$ .

| Condition               | Idle RMS (mm) |           | $F_{\max}$ (N) |           |
|-------------------------|---------------|-----------|----------------|-----------|
|                         | Mean          | $\pm$ Std | Mean           | $\pm$ Std |
| Clean, 0 ms delay       | 0.42          | 0.03      | 96             | 1         |
| Clean, 10 ms delay      | 2.30          | 0.16      | 47             | 2         |
| Clean, 30 ms delay      | 2.62          | 0.22      | 46             | 3         |
| Clean, 50 ms delay      | 2.61          | 0.42      | 49             | 8         |
| Clean, 100 ms delay     | 2.23          | 0.25      | 48             | 4         |
| Clean, 150 ms delay     | 2.40          | 0.22      | 45             | 1         |
| Low noise, 0 ms delay   | 6.33          | 2.57      | 96             | 1         |
| Low noise, 10 ms delay  | 12.82         | 3.93      | 47             | 2         |
| Low noise, 30 ms delay  | 7.31          | 2.51      | 50             | 1         |
| Med noise, 0 ms delay   | 34.00         | 11.28     | 94             | 3         |
| Med noise, 10 ms delay  | 41.66         | 23.24     | 45             | 10        |
| Med noise, 30 ms delay  | 35.10         | 16.83     | 41             | 10        |
| High noise, 0 ms delay  | 3.00          | 0.55      | 95             | 3         |
| High noise, 10 ms delay | 3.00          | 0.55      | 95             | 3         |
| High noise, 30 ms delay | — (fell)      | — (fell)  | — (fell)       | — (fell)  |

Noise (1- $\sigma$ ): low = 0.01 rad/s (ang. vel.), 0.01 m/s<sup>2</sup> (grav.), 0.001 rad (joint pos.); med = 5 $\times$  low; high = 10 $\times$  low. Two distinct delay regimes: **performance** bifurcation at  $\sim 10$  ms ( $F_{\max}$ : 96 N  $\rightarrow$   $\sim 47$  N); **stability** threshold >150 ms (zero falls at all tested clean delays,  $N=5$ ; post-bifurcation idle  $\approx 2.5$  mm,  $F_{\max}\approx 45-49$  N). The apparent medium $\rightarrow$ high noise improvement (34 mm $\rightarrow$ 3 mm) occurs because high noise holds EMA above threshold, disengaging the anchor into P-only mode. Non-monotonic low-noise RMS (12.82 mm at 10 ms vs. 7.31 mm at 30 ms) reflects resonance: 10 ms phase-shifts the anchor into intermittent engage/release; at 30 ms EMA stays above threshold (medium noise follows the same pattern).

integral gain  $K_i$  require minimal adjustment across platforms; the critical parameter is  $\tau_{\text{release}}$ .

**Gate failure modes.** Stuck-at-1 ( $g_{\text{anchor}}\equiv 1$ ) causes windup  $\rightarrow$  fall (cf. S4); stuck-at-0 yields P-only limit cycle (cf. L0); stuck-half-open triggers parametric resonance (cf. S5,  $F_{\min}=9$  N). All three map to measured ablation rows (Table IV), providing implicit FMEA coverage.

**Torque rate-limit invariance.** ACC is **invariant** across 200 Nm/s to 800 Nm/s (4 $\times$ )—its ceiling is architectural, not actuator-limited.

**Compound-disturbance recovery.** Scenario A (push during 0.65 m $\rightarrow$ 0.50 m squat): survival degrades 84% (94 N $\rightarrow$ 15 N, 95%CI [11, 19] N,  $N=5$ ). Scenario B (sequential 90 N forward  $\rightarrow$  60 N backward): **5/5 survival** (95%CI [0.48, 1.0]), ring-down <1 s.

## VI. DISCUSSION AND CONCLUSION

ACC achieves all demonstrated behaviors without neural network training, is hypothesized to provide a *reduced* sim-to-real transfer gap vs. learned policies—pending hardware validation [22]. Each gate was added after tracing a measured failure mode to its root cause: (i) bistable envelope; (ii) global-I windup; (iii) parametric pumping; (iv) hard-leash oscillation; (v) self-referential stiffness. All thresholds frozen prior to the ablation sweep (Table IV).

**Statistical power.** Welch’s  $t$ -test, Bonferroni-adj.  $\alpha=0.007$  (7 comparisons); L0 $\rightarrow$ L1 and L2 $\rightarrow$ L3 significant ( $p<0.001$ , Cohen’s  $d\approx 5.3, 6.4$ ). Primary claims ( $\sim 53\times$  idle precision,

S5 collapse) are unambiguous. Flight/terrain:  $N=10$ , Clopper–Pearson 95%CI [0.69, 1.00].

**Limitations.** (1) Backward push is weakest ( $F_{\min}$  drops  $\sim 50\%$  vs. median), inherent to fore/aft geometry. (2) Oblique  $45^\circ$  ledge descent at  $\geq 20$  cm exceeds lateral balance authority. (3) All results are simulation-only (MuJoCo); hardware validation remains future work. (4) ACC exhibits two distinct delay thresholds: a **performance** bifurcation at  $\sim 10$  ms ( $F_{\max}$  drops  $\sim 95$  N  $\rightarrow$   $\sim 45$  N, the 0.5 ms end-to-end budget margin is for this threshold); and a **stability** threshold  $> 150$  ms (robot survives all tested clean delays, zero falls,  $N=5$ ; see Deployment Risks). A jitter spike exceeding the performance threshold degrades push recovery temporarily but does not cause a fall—ACC’s gates depend on physical state, not tight timing. (5) ACC operates at 100 Hz vs. 50 Hz for LQR/PPO baselines; the  $2\times$  bandwidth advantage contributes partially to performance. A 50 Hz ACC evaluation with recomputed discrete coefficients ( $\alpha_a \approx 0.58$ ,  $\alpha_r \approx 0.013$ , anchor leak 0.010) achieved 0.05 mm idle RMS—preserving sub-millimeter precision at the baseline rate, confirming the architectural contribution is not rate-dependent for steady-state regulation—while push recovery degraded below 30 N (vs. 83 N at 100 Hz), confirming that disturbance rejection benefits from the 100 Hz rate. DT-LQR, which eliminates the PID servo path (the dominant lag source), still falls at 0.72 s at 50 Hz without any push; ACC at 50 Hz stands indefinitely. Together, these results establish that the architecture determines the qualitative outcome (stand vs. fall) while the rate contributes quantitative improvement to dynamic recovery margins. (6) Push survival degrades 84 % during height transitions (shared torque budget). (7) Six classical baselines—four 4-state TWIP variants (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo and direct-torque)—all fail (100 %). A full-state LQR from finite-difference linearization of the complete 10-DOF MuJoCo plant was attempted but produced unreliable gain matrices due to ill-conditioned contact dynamics at the standing equilibrium ( $\kappa(A) \approx 10^{10}$ ); robust linearization of articulated-leg contact dynamics at unstable equilibria is an open challenge. The hip-roll coupling  $b_{r,h} = -5.0$  in the 6-state LQR is empirical; a sensitivity sweep is deferred. (8) Architectural generality across morphologies remains to be demonstrated. (9) Push ablation at  $N=10$  has min. detectable effect  $\sim 3.2$  N; sub-2 N differences non-significant (see Statistical power). The PPO baseline (5.4M steps, single seed) may not be fully converged; the 85 % fall rate is reported as observed, not as an upper bound on RL performance for this platform. A task-priority QP whole-body controller was benchmarked both standalone and as an assist on ACC and did not resolve the precision–recovery trade-off (§V-E); re-running with known implementation bugs fixed is deferred (the architectural argument—absence of regime-gating in a fixed-task-priority formulation—does not depend on it). A tuned convex MPC on the full 10-DOF plant remains deferred because finite-difference linearization at the standing equilibrium is ill-conditioned ( $\kappa(A) \approx 10^{10}$ , Limitation 7), so robust full-plant MPC is an open prerequisite rather than a missing comparison.

**Deployment Risks and Mitigations.** ACC uses raw torque, 400 Nm/s rate limit, standard sensors (IMU + encoders). We distinguish two latency thresholds: a **performance** threshold ( $\sim 10$  ms) where  $F_{\max}$  bifurcates 96 N  $\rightarrow$   $\sim 47$  N, and a **stability** threshold ( $> 150$  ms, clean sensors,  $N=5$ , zero falls) where the robot actually falls. ACC’s gates (proximity, envelope, pitch) depend on physical state, not timing precision—the architecture is intrinsically delay-tolerant for stability. The 150 ms+ stability threshold is consistent across all physical gating mechanisms: the proximity gate (spatial, 15 cm), asymmetric envelope (temporal,  $\tau_r \approx 1.5$  s), and damping boost are all state-driven rather than schedule-driven.

Estimated non-RTOS worst-case budget: sensor readout  $\sim 1$  ms + state estimation  $\sim 2$  ms + ACC computation  $\sim 3$  ms + actuation latency  $\sim 3.5$  ms =  $\approx 9.5$  ms end-to-end. The 0.5 ms margin below the 10 ms performance threshold means any single jitter source (OS scheduling preemption, bus contention) can degrade push recovery margin temporarily; the stability margin remains  $> 140$  ms. On a real-time OS with a dedicated balance-loop microcontroller (e.g., 1 kHz RT Linux or bare-metal MCU), jitter is typically  $< 0.1$  ms, recovering 2 ms to 4 ms of budget and making the performance margin comfortable. Neither scenario has been validated on hardware—latency characterization with an oscilloscope and jitter histogram is required before deployment. The discarded WBC-assist variant (§V-E) is excluded from the hardware path not only because it adds no measurable balance benefit (225-scenario equivalence), but because its QP solve alone (p99 3.43 ms) exceeds the 0.5 ms margin below the 10 ms performance threshold by  $6.9\times$  in the offline path; JAX-JIT incremental integration was prototyped but not validated, and the zero-net benefit makes further optimization moot. Hardware degradation: 0.73 mm  $\rightarrow$  projected 1 – 3 mm (encoder quantization, backlash, stiction, tire compliance, IMU noise; cf. Table VII). Staged commissioning: static torque  $\rightarrow$  standing  $\rightarrow$  push recovery  $\rightarrow$  flight (tethered)  $\rightarrow$  terrain.

We presented ACC, a two-channel torque assembly where a PD balance core is augmented by a proximity-gated anchor integral achieving  $0.73 \pm 0.07$  mm idle CoM RMS ( $N=10$ , 95%CI [0.68, 0.78] mm) with omnidirectional push survival ( $F_{\min}=83$  N,  $F_{\text{med}}=117$  N). Six classical baselines—four 4-state TWIP variants (LQR, LQR+AW, LQR-DT, PI+AW) and two 6-state coupled variants (servo and direct-torque)—all fail (100 %); adding integral action to LQR-optimal feedback does not prevent the fall, confirming ACC’s structured prior as the essential contribution. ACC tolerates moderate noise, is rate-limit-invariant (200 Nm/s to 800 Nm/s), exhibits a push-performance delay bifurcation at  $\sim 10$  ms with stability maintained beyond 150 ms, and handles contact loss and terrain without neural network training. Future work: hardware validation, EMA settle-time characterization, and extending conditional activation to locomotion and as a structured prior for residual RL.

## APPENDIX A. WHY SATURATION-TRIGGERED AW IS INSUFFICIENT

Saturation-triggered AW triggers on actuator saturation; on this platform, position error grows to 10—30 cm during a push while wheels stay within their 20 rad/s range, so saturation-triggered AW never activates. A static dead-zone either over-restricts or under-restricts; ACC’s proximity gate with continuous smoothstep avoids both. State-gated conditional integration is practiced in process control; ACC couples spatial gating with asymmetric envelope-detected damping, providing temporal gating absent in static-threshold schemes. AW-augmented baselines (LQR+AW, PI+AW) are evaluated in Table VI; broader AW taxonomy comparison is deferred. Code and parameters at <https://github.com/thuong180702/Wheeled-bipedal-robot-simulation> (commit f92640c).

## APPENDIX B. WBC BASELINE IMPLEMENTATION DETAILS

### B.1 Standalone WBC (W1) — Full Metric Table

Table VIII reports all measured metrics for the standalone QP-WBC (W1) vs. ACC, from the 225-scenario three-arm counterfactual evaluation and the 4-scenario Phase 3D full-batch execution. The WBC was evaluated with known implementation bugs (Table IX); W1 numbers may therefore be pessimistic. However, the failure mechanism—velocity-damping posture, single-linearization, no rolling model—is architectural, and the absence of regime-gating (the core ACC contribution) is structural, not bug-dependent.

TABLE VIII  
STANDALONE QP-WBC (W1) VS. ACC — FULL METRIC COMPARISON

| Metric                               | WBC-only | ACC (V3) | Ratio (WBC/ACC) |
|--------------------------------------|----------|----------|-----------------|
| Total falls (225 scenarios)          | 5,985    | 347      | 17.2×           |
| Single-push falls                    | 3,353    | 118      | 28.4×           |
| Falls (Phase 3D, 4 step_e scenarios) | 14,713   | 0        | —               |
| Survival time at 0.60 m (s)          | 0.52     | 20.0     | 0.026×          |
| Pitch RMS (°)                        | 41.8     | 15.9     | 2.63×           |
| Roll RMS (°)                         | 96.8     | 84.2     | 1.15×           |
| Yaw drift (°)                        | 8.0      | 4.24     | 1.89×           |
| Height RMS (m)                       | 0.253    | 0.261    | 0.97×           |
| Planar drift (m)                     | 2.57     | 2.42     | 1.06×           |
| Posture error (ratio vs. ACC)        | —        | —        | 8.05×           |
| Yaw error (ratio vs. ACC)            | —        | —        | 15.43×          |
| QP solver success rate               | 85.85 %  | —        | —               |

Sources: 225-scenario batch (see `V3_vs_V3_Assist_comparison_report.md`); Phase 3D (full\_batch\_verdict.json); survival time (best\_torque\_wbc\_summary.json). All runs with ACC profile, no gain retuning. WBC evaluated with known bugs (Table IX); W1 numbers may be pessimistic.

### B.2 Known WBC Implementation Bugs (F1–F4)

Table IX lists the four critical bugs identified during post-hoc audit of the WBC implementation. These bugs affect W1 (standalone) metrics; W2 (assist) zero-net is bug-independent because the gate blocks WBC torque before it is computed.

TABLE IX  
WBC IMPLEMENTATION BUGS (POST-HOC AUDIT)

| ID | Description                                                                                   | Effect on Metrics                                                                                                         |
|----|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| F1 | Contact Jacobian zero-gradient: leg joint columns = 0 (autodiff through cached FK).           | Ground reaction forces produce zero moment on leg joints; $\tau_{wbc}$ meaningless at hip-pitch/knee ( $K_{role}=0.60$ ). |
| F2 | Fast solver hardcodes <code>feasibility_only</code> —drops entire task stack.                 | QP finds minimum-norm $\ddot{q}$ ; all task modes produce identical output.                                               |
| F3 | Offline eval passes base-z (0.536 m) as height cmd. instead of CoM-z (0.404 m)—13.2 cm error. | ACC gain schedule and feedforward at wrong operating point; drift/heading/centering gates disabled.                       |
| F4 | Gate height conflict: WBC at 0.67 m, V3 commands CoM at 0.45—0.60 m.                          | $q_{height} \approx 0$ at all test heights; WBC gate forced closed.                                                       |

Source:

`docs/validation/v3_wbc_assist_root_cause_audit.md`.  
F1–F4 affect W1 (standalone); architectural argument (absence of regime-gating) does not depend on them. W2 (assist) zero-net is bug-independent: gate blocks WBC before torque is computed.

### B.3 W2 Zero-Net is Bug-Independent

The assist gate  $\alpha_j$  is computed from physical state (pitch, roll, height deviation, push magnitude, divergence) before WBC torque is evaluated. Since  $\alpha_j \approx 0$  in  $\geq 99\%$  of steps, the WBC’s internal correctness—buggy or not—is irrelevant to the assist outcome. The 225-scenario equivalence (ACC + WBC assist  $\equiv$  ACC within measurement noise) is therefore a robust result. The keyframe test (gates manually overridden) reveals the torque-blend conflict (roll +35 %) independently of F1–F4: with the gate forced open, WBC torque is applied, and the decentralized-vs-centralized conflict manifests as roll degradation.

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